Fiscal Year:	FY 2005	**Task Last Updated:**	FY 01/06/2005
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Project Title:	Investigations of the Influence of Air-liquid Interfacial Stresses on Pulmonary Epithelial Cells in a Microgravity Environment		
Division Name:	Physical Sciences		
Program/Discipline:	FLUID PHYSICS		
Program/Discipline--Element/Subdiscipline:	FLUID PHYSICS--Biofluids		
Joint Agency Name:		**TechPort:**	No
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Space Biology Element:	None		
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Comments:			
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Task Description:	Pulmonary airway closure is a potentially dangerous event that can occur in microgravity environments and may result in limited gas exchange for flight crew during long-term space flight. To reopen collapsed airways, a semi-infinite bubble of air must separate airway walls that are held in apposition by a viscous lining fluid. However, the fluid mechanical stresses exerted by the migrating bubble can be very large and damage the lung. The fluid mechanics issues related to these studies are complex and are intimately related to the physiological responses of the lung. We will investigate the cellular responses to interfacial and viscous stresses that occur as a surfactant-doped bubble progresses over pulmonary epithelial cells that line the airways and alveoli in the lung. By identifying the magnitude of these stresses and the physiological response to the stimulus, we will provide insight of appropriate methods for opening collapsed airways, and how these methods depend upon the physicochemical properties of the airway lining fluid. As such, our studies will develop the link between fluid physics and potentially hazardous life-science issues that may be manifested during long-term space flight. Specific Aim #1: Study the fluid dynamical and surfactant physicochemical responses related to the opening of collapsed pulmonary airways in a microgravity environment. Specific Aim #2: Examine cellular trauma due to bubble progression in a physical model of a collapsed airway. Specific Aim #3: Predict flow and surfactant behavior that will minimize cellular injury during the opening of collapsed airways. Our studies are of importance to NASA for two reasons. First, pulmonary airway closure is more ubiquitous in a microgravity environment than on earth - this has been demonstrated in microgravity life-science studies of pulmonary function during short-term parabolic flights and long-term investigations on the SpaceLab (SLS-1). In microgravity, airway closure exists throughout the lung, rather than only in the basal 'gravitationally dependent'regions. While this closure occurs primarily at low lung volumes, recent parabolic flight studies of inhaled gas boluses indicate that airway closure in microgravity occurs in close proximity to airways that remain open, and may provide the inhomogeneity in the lung. Such closure may impact pulmonary function in microgravity environments during 'normal' and disease conditions. It is thus important to understand the complex fluid-structure interaction of this system to determine the likelihood of pulmonary cellular damage in microgravity,and to identify methods of reducing the effects of these stresses on pulmonary cells. In addition, airway closure occurs in disease, and thus further understanding of the interaction between air-liquid interface behavior and pulmonary cellular responses could lead to advanced therapeutics for individuals on earth. The small physical size of the airway/bubble system that exists in regions of airway closure means that gravitational effects are small. While flight studies of this system are not proposed herein, we may in the future find it important to scale-up this system so as to measure the convection, transport and stress behavior in a system of larger physical size. Such studies would have to be performed in a microgravity environment to remove the effects of gravity. However, it is important to develop a basic understanding of these systems in a terrestrial environment so that we can determine the scaling principles from which the microgravity flight studies would be based.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	
Task Progress:	Specific Aim #1: Study the fluid dynamical and surfactant physicochemical responses related to the opening of collapsed pulmonary airways in a microgravity environment. We have experimentally investigated oscillating flow in an occluded rigid airway, and have identified parameter ranges (frequency and amplitude) that result in increased surfactant transport. Our studies indicate that low amplitude, high-frequency oscillations are most beneficial to the additional transport of surfactant to the bubble tip. This suggests that a high-frequency 'jitter' may be useful in ventilation scenarios for opening occluded airways. Specific Aim #2: Examine cellular trauma due to bubble progression in a physical model of a collapsed airway. We have completed studies of cellular responses to bubble progression in a Hele-Shaw cell. The results of these studies have been correlated to computational investigations of Specific Aim #1, and indicate that the cellular damage is due to normal stress gradients (variation in stress perpendicular to the cell layer) that are developed as a bubble progresses over the cell layer. Increasing the bubble velocity decreases the stresses, and thus can be used to protect the cells from damage. Recent studies indicate that the damage is due to the intensity of stress, and not the duration (to leading order). Specific Aim #3: Predict flow and surfactant behavior that will minimize cellular injury during the opening of collapsed airways. Computational models have been developed of both steady and oscillating flow in an occluded rigid airway, and have used these models to assess surfactant transport characteristics. These studies show that oscillation of the interface can cause a beneficial buildup of surfactant in the bubble tip region. Furthermore, the development of multilayers can be advantageous in the development of low surface tension regions that will protect the lung from abnormal stresses. Unsteady reopening simulations indicate the existence of a viscous stick-slip response that might be of importance to understanding rapid 'avalanche' type reopening that has been observed in-vivo.
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